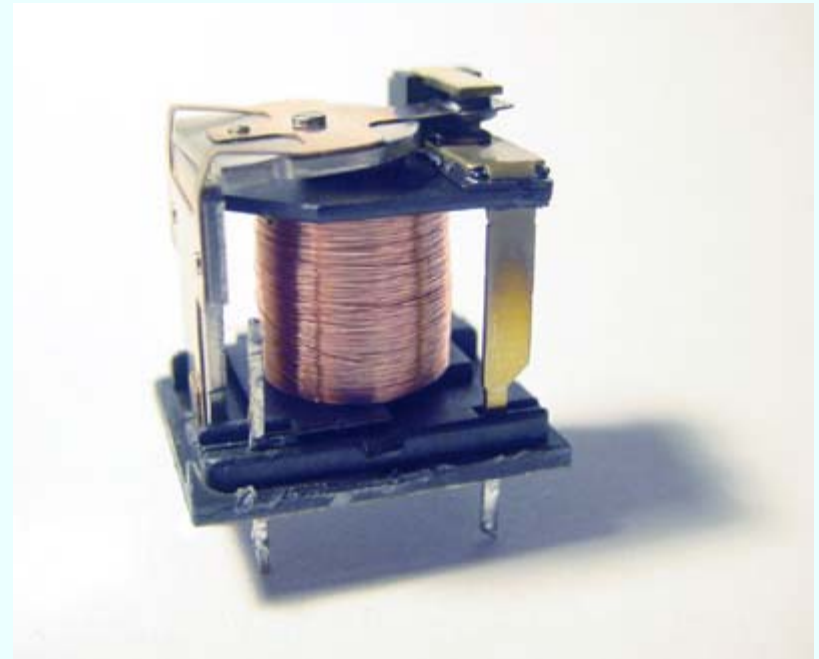
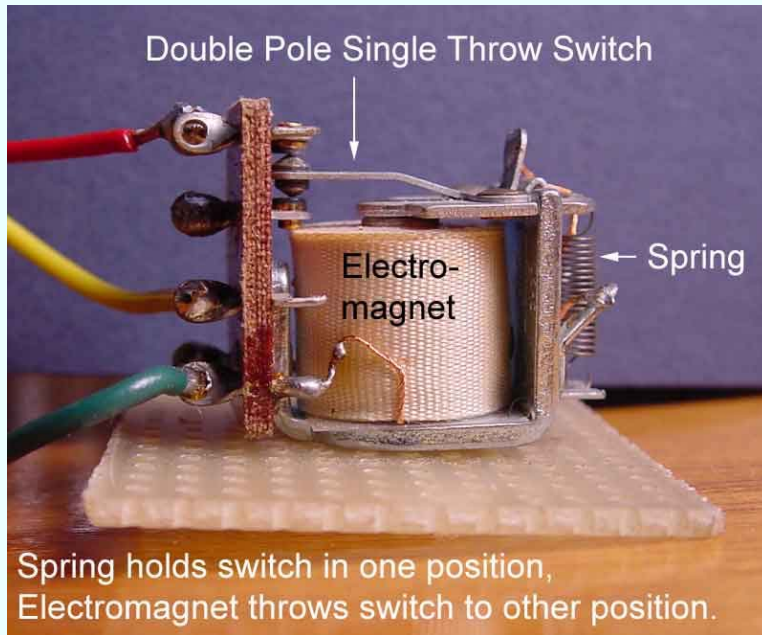


Relay - a very simple device



A relay is a simple electromechanical switch made up of an electromagnet and a set of contacts.

Relay - what is it good for

In general, the point of a relay is to use a small amount of power in the electromagnet - coming, say, from a small dashboard switch or a low-power electronic circuit - to move an armature that is able to switch a much larger amount of power.

E.g. You can use 1W to switch 10000W (10kW).

Additionally, thanks to auxiliary contacts (normally opened or normally closed) can be used to implement additional logic.

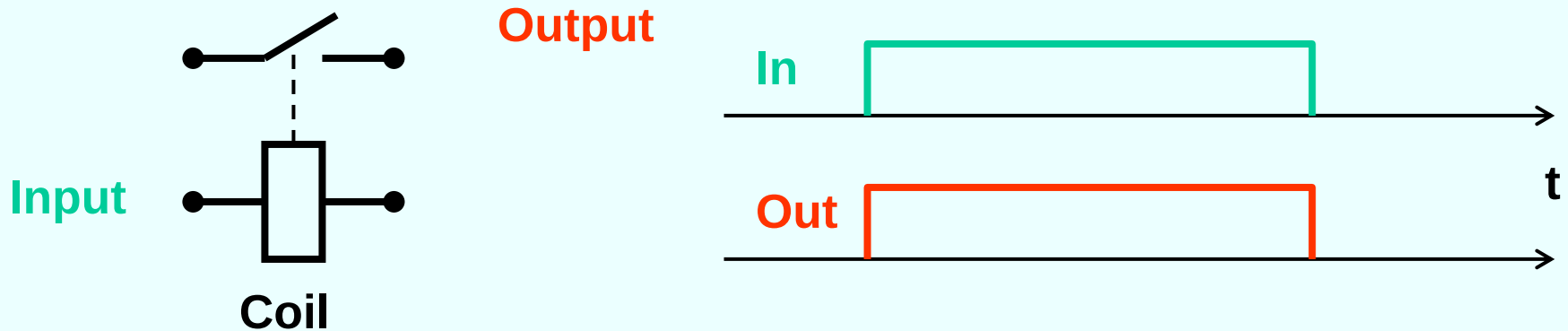
Coil excited – boolean value „1”

Coil not excited – boolean value „0”

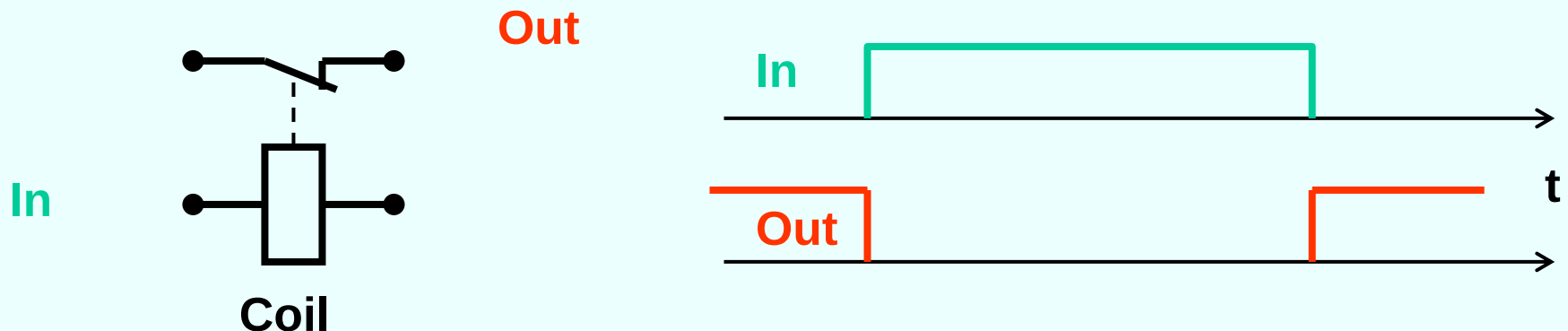
Opened contact – value „0”

Closed contact – value „1”

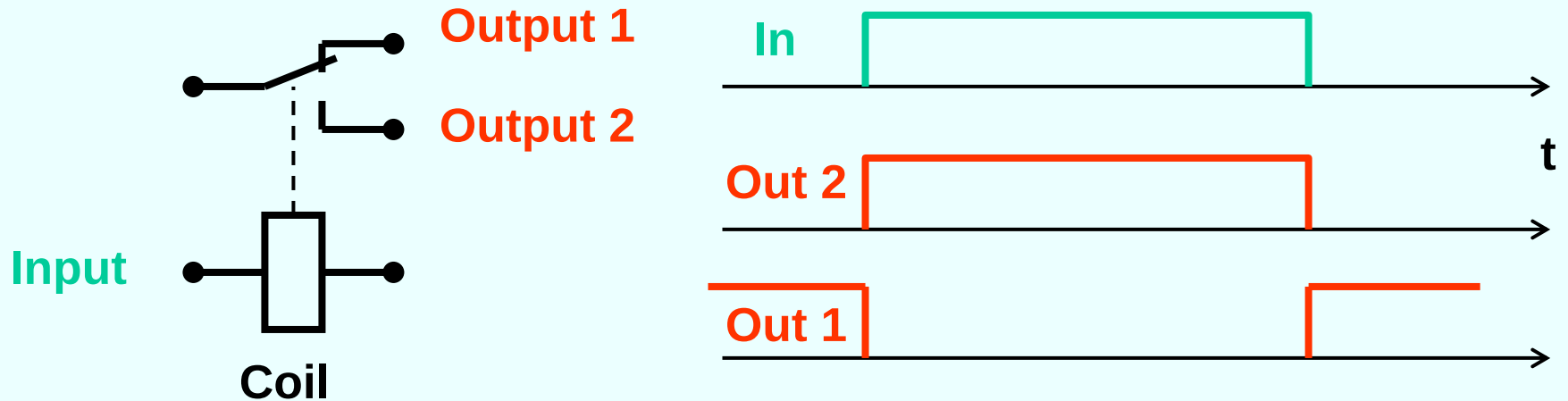
Relay with normally open[ed] contact NO



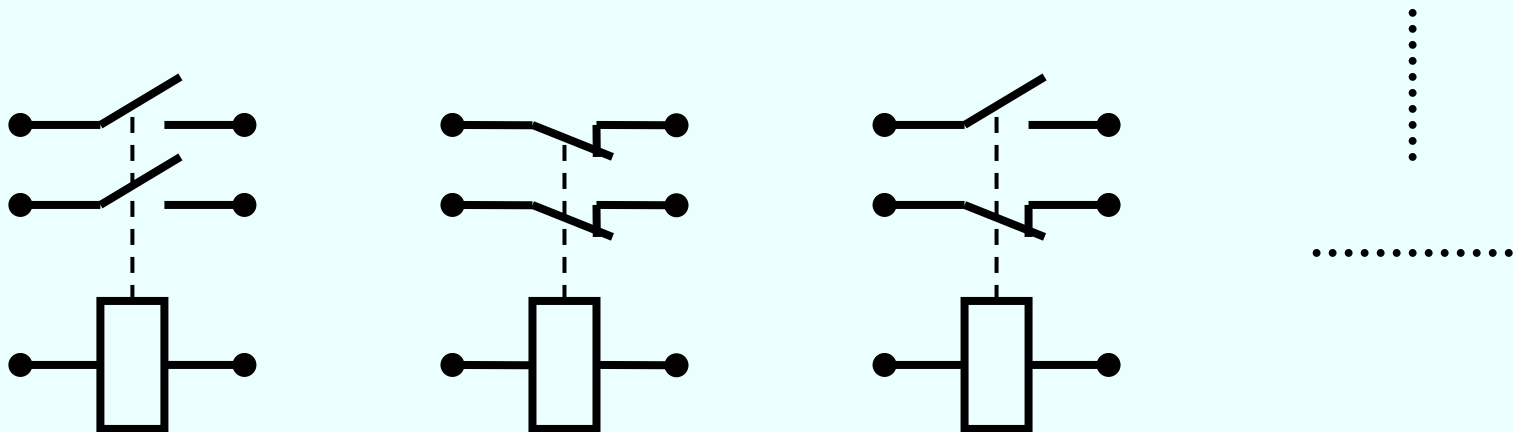
Relay with normally closed contact NC



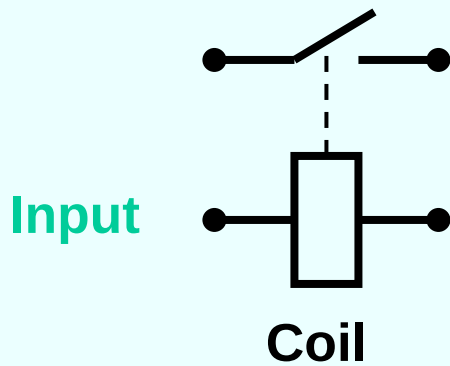
Relay with NO and NC contacts



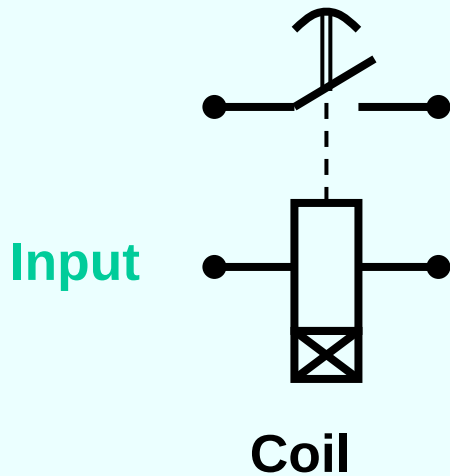
Relay with multiple NO and NC contacts



„Normal” relay



Time-delay relay (normally-open, time-closed)

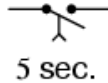


ΔT – sec, min, hours

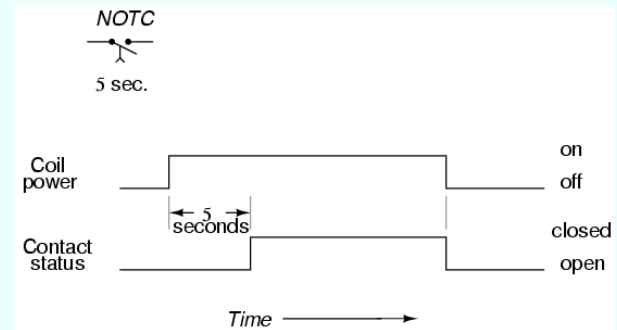
Time-delay relays (four basic devices)

NOTC

Normally-open, timed-closed

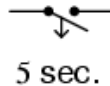


*Closes 5 seconds after coil energization
Opens immediately upon coil de-energization*

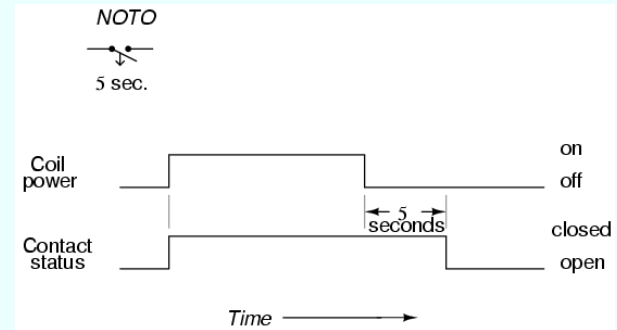


NOTO

Normally-open, timed-open

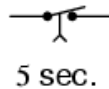


*Closes immediately upon coil energization
Opens 5 seconds after coil de-energization*

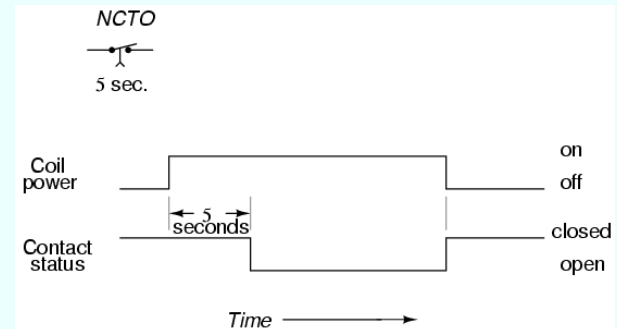


NCTO

Normally-closed, timed-open

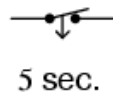


*Opens 5 seconds after coil energization
Closes immediately upon coil de-energization*

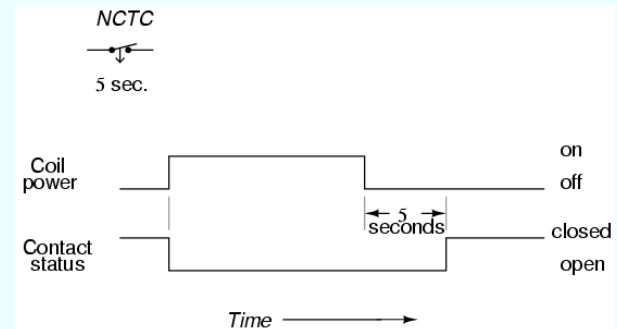


NCTC

Normally-closed, timed-closed



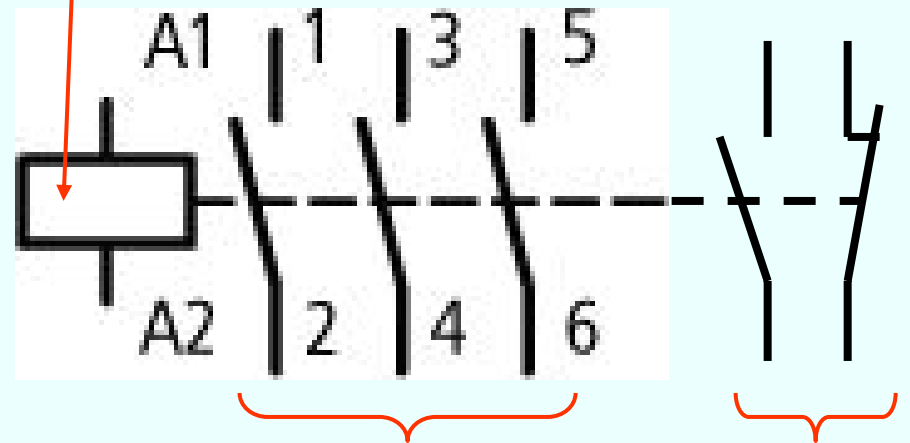
*Opens immediately upon coil energization
Closes 5 seconds after coil de-energization*



Relay (an example)



Coil



Main contacts

Auxiliary contacts

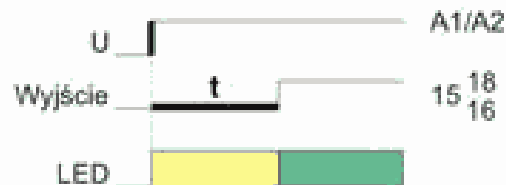
Time-delay relay (an example)

saia-burgess

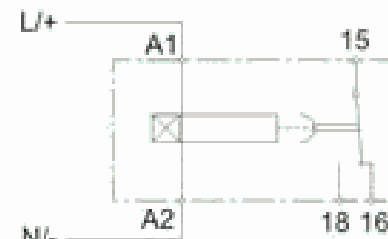


Napięcie zasilania 24...240VAC, 24...48VDC
 Zakres czasowy 0.05s...10h
 6 podzakresów czasowych
 1s, 10s, 1min, 10min, 1h, 10h
 4 funkcje czasowe
 11, 12, 21, 42
 Rodzaj wyjścia 1P
 Zdolność łączeniowa $I_{th}=5A/U=250VAC$
 Szerokość obudowy 17.5mm (1 moduł)
 Montaż na szynie 35mm
 Certyfikaty B, CE, GL

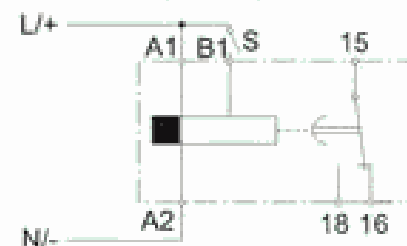
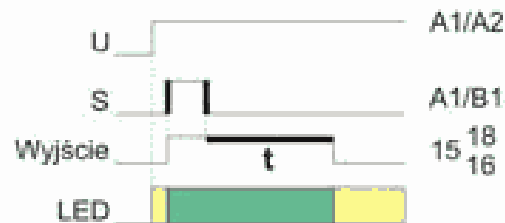
"11" - opóźnione załączanie



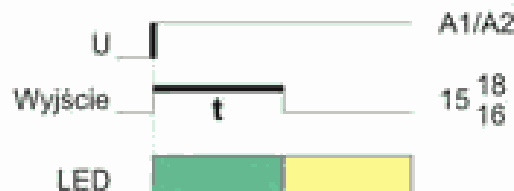
On Delay



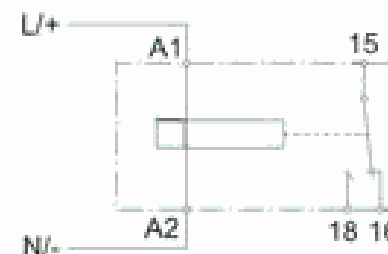
"12" - opóźnione rozłączanie (wyzw...: zewn. impuls sterujący)



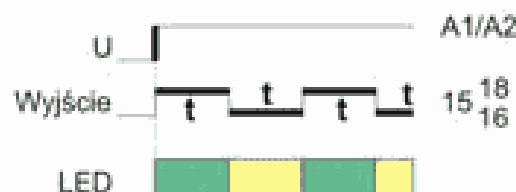
"21" - opóźnione rozłączanie



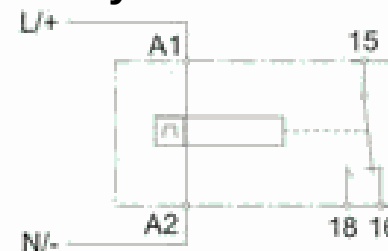
Off Delay



"42" - impulsator symetryczny



Recycle



Exercise 1

You've got:

2 relays with 3-f main contacts and with as many auxiliary contacts as you want :)

2 push button switches NO (green ones)

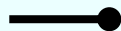
1 push button switch NC (red one)

1 induction motor

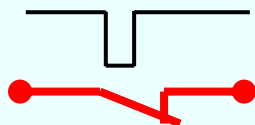
Your goal is to operate drive in two directions safely (without a risk of short circuiting the mains)

Let the competition begin!

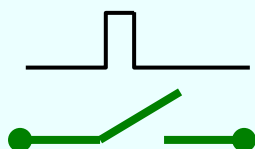
24V



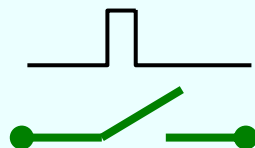
Stop



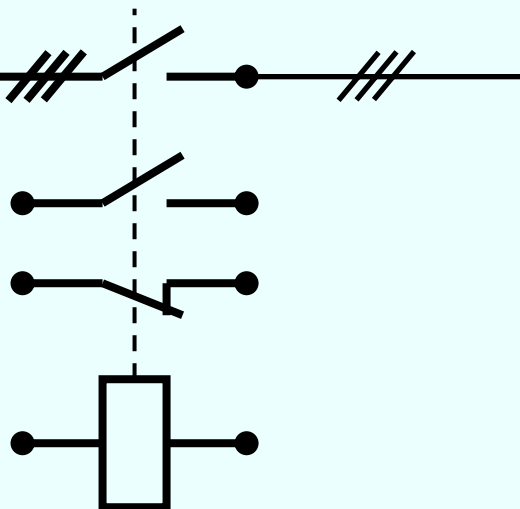
Start left



Start left

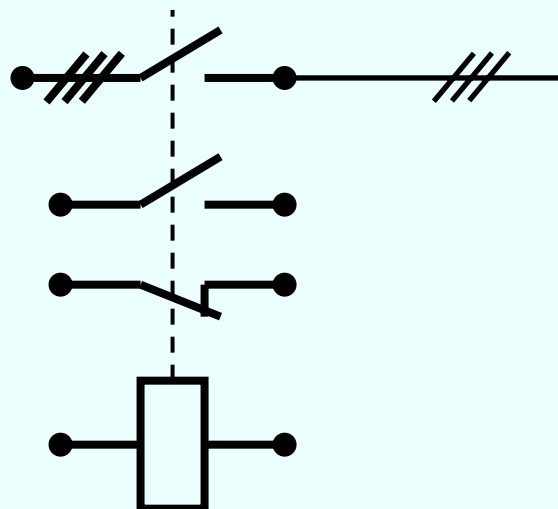


A
B
C



Right (clockwise)

B
A
C



Left (counter-clockwise)

Induction
motor

24V

Stop

Start right

Start left

?

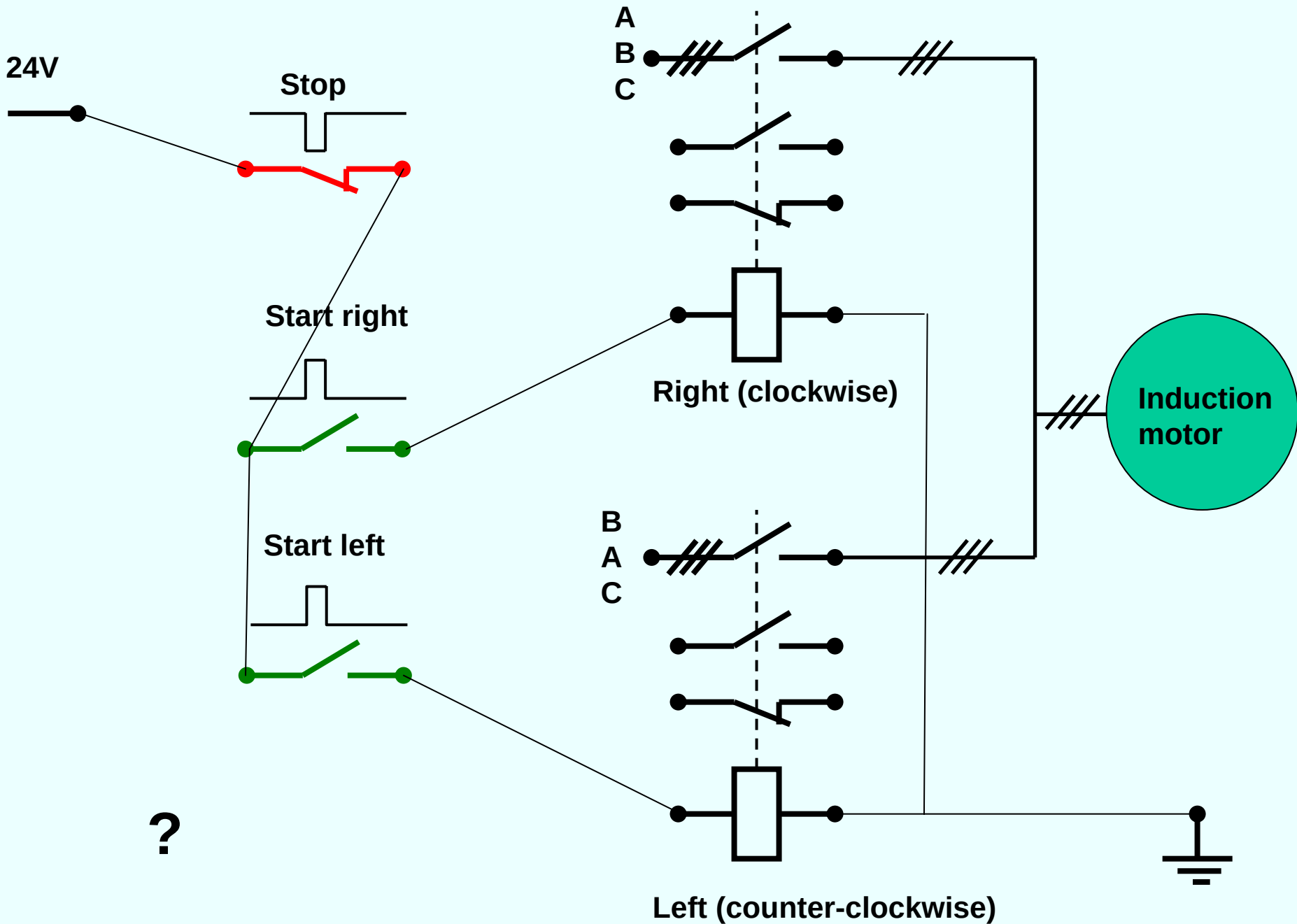
A
B
C

Right (clockwise)

B
A
C

Left (counter-clockwise)

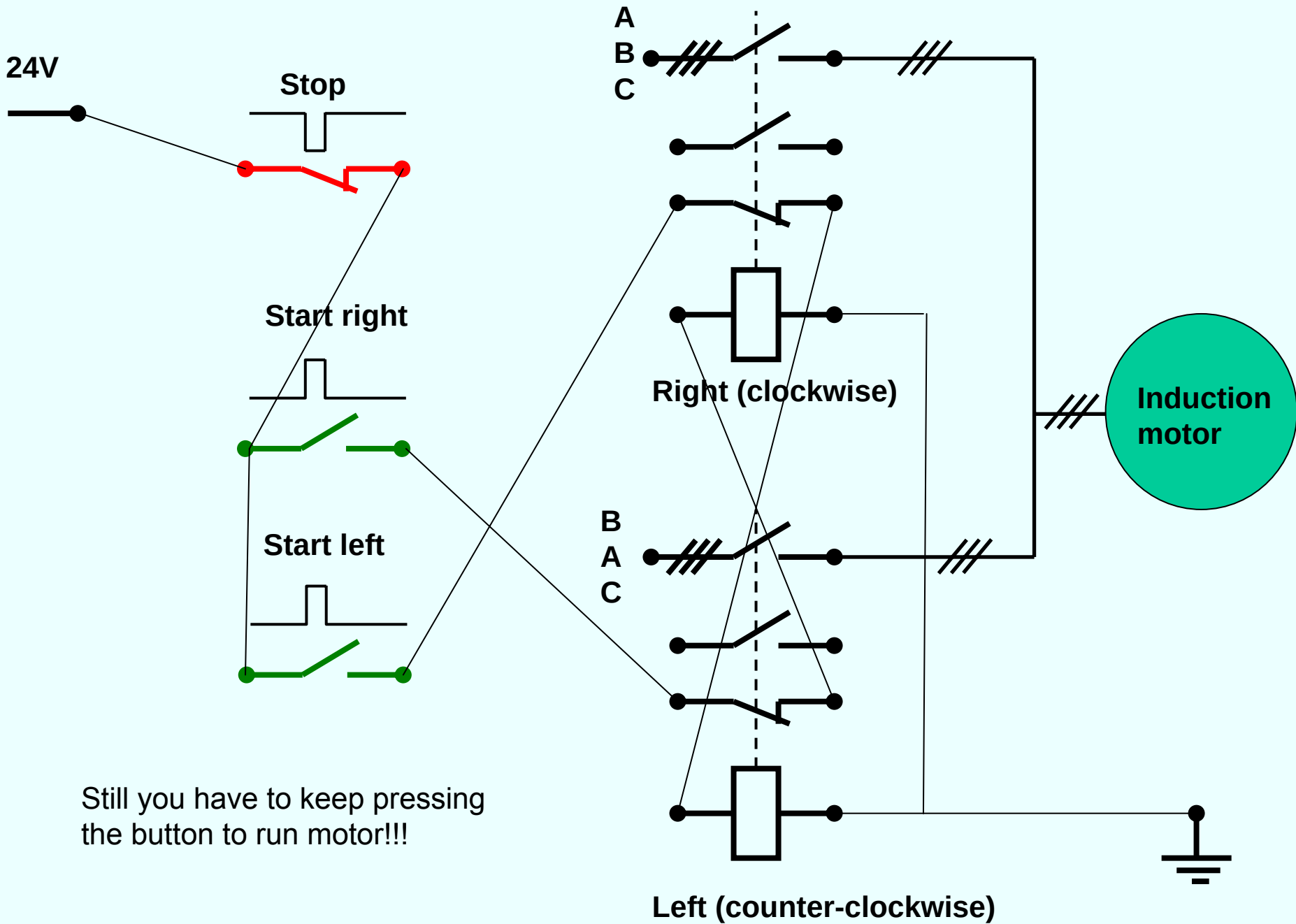
Induction
motor

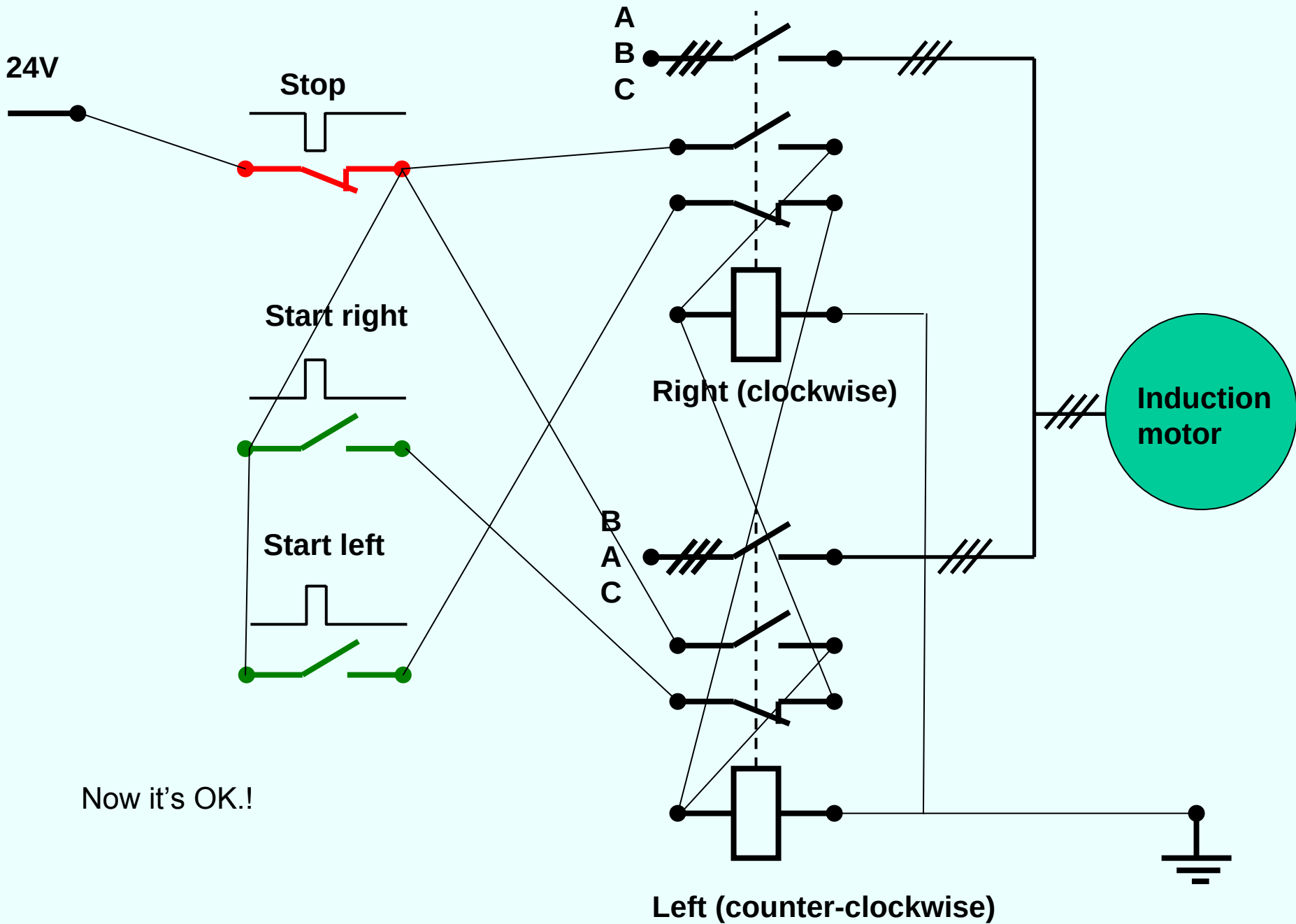


There are 2 errors:

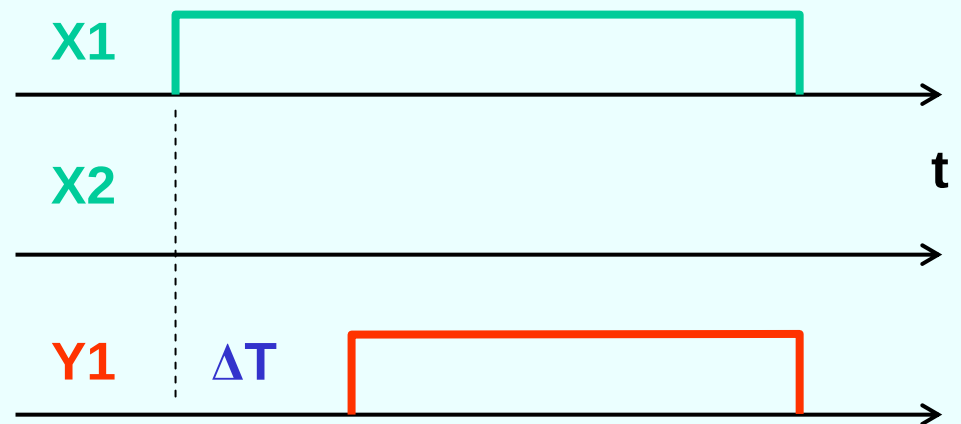
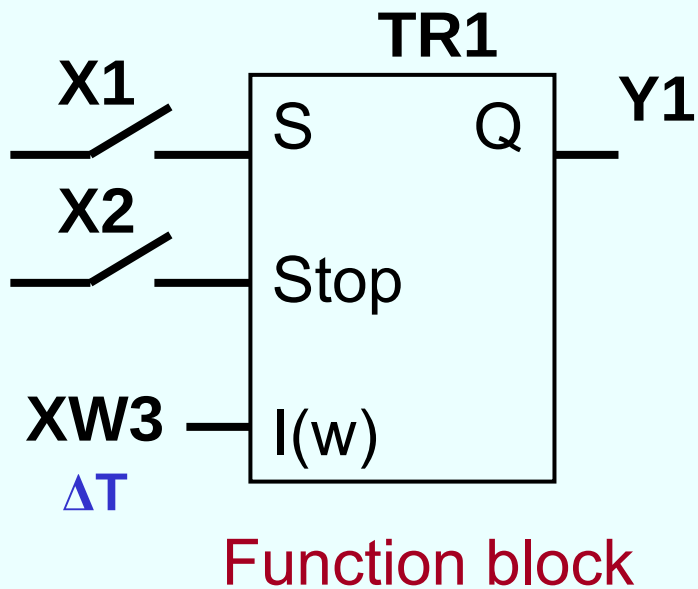
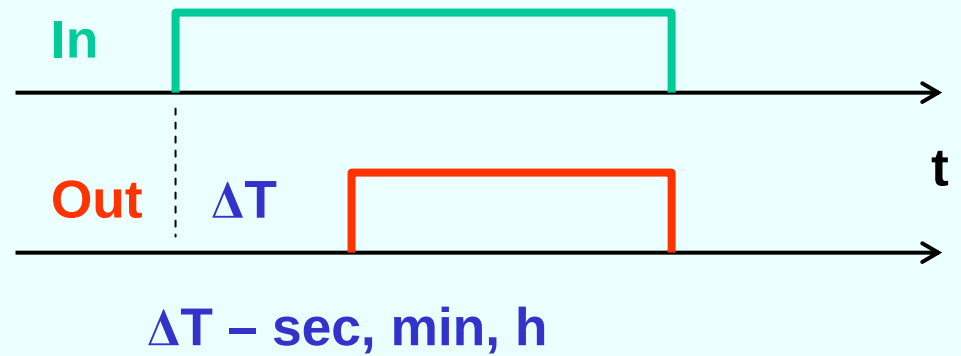
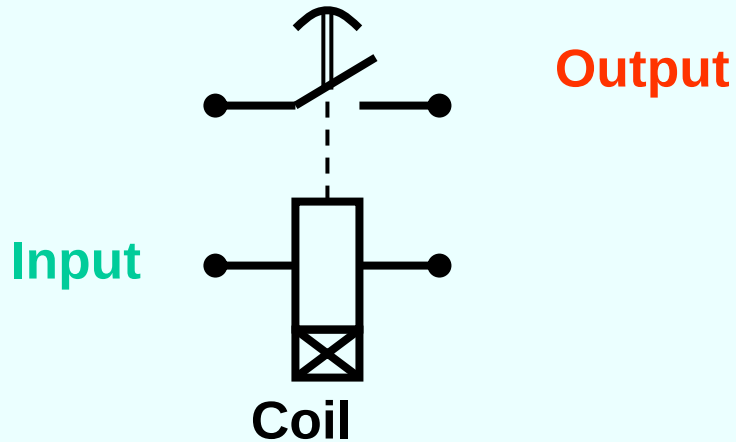
- 1. If someone presses both green buttons, the mains is short circuited!**
- 2. The voltage is not sustained at motor terminals - you have to keep pressing the button to run motor!!!**

Let's solve the first one...



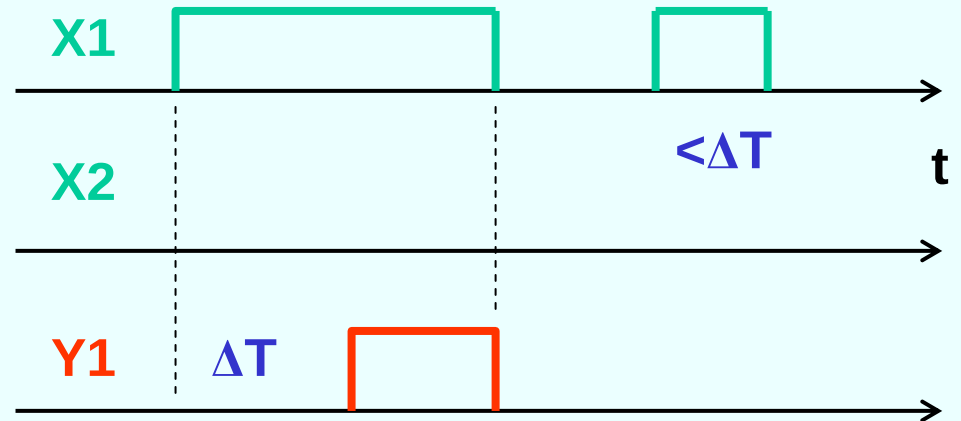
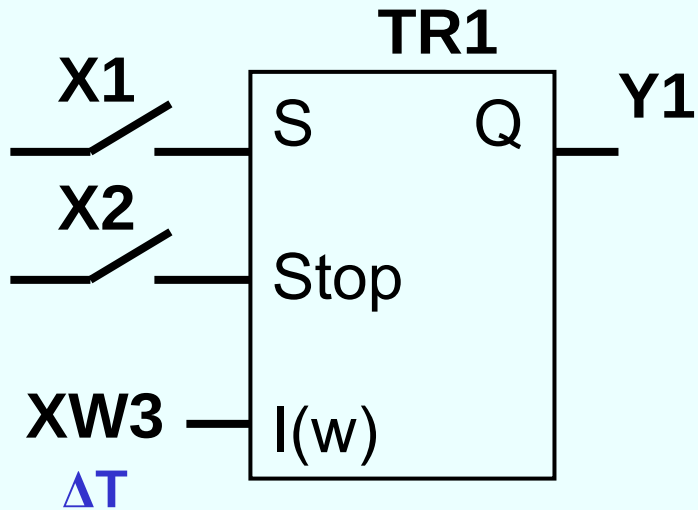


On delay relay



$X \Rightarrow I, M, Q$ $Y \Rightarrow M, Q$

$XW \Rightarrow MW, IW, KW$ $\times 100 \text{ ms}$



List of instructions

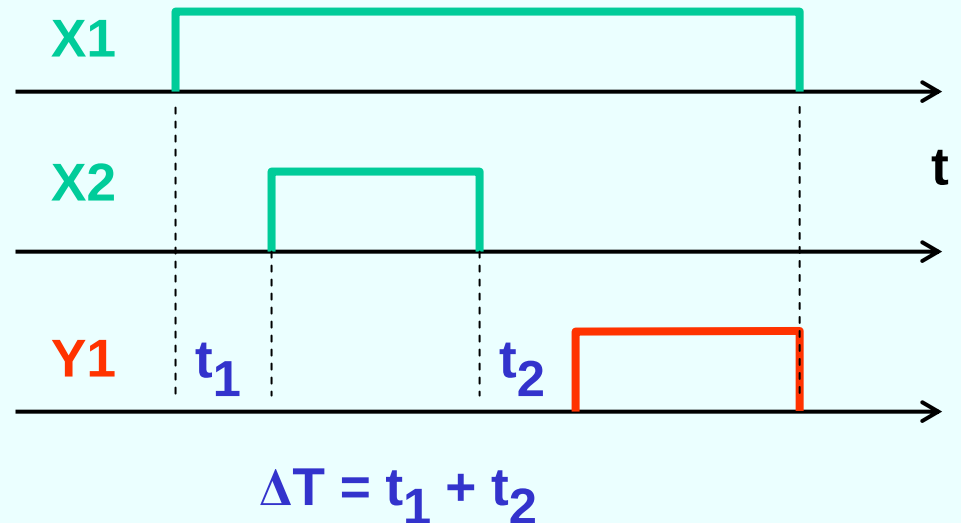
TR1

[]S: **X1**

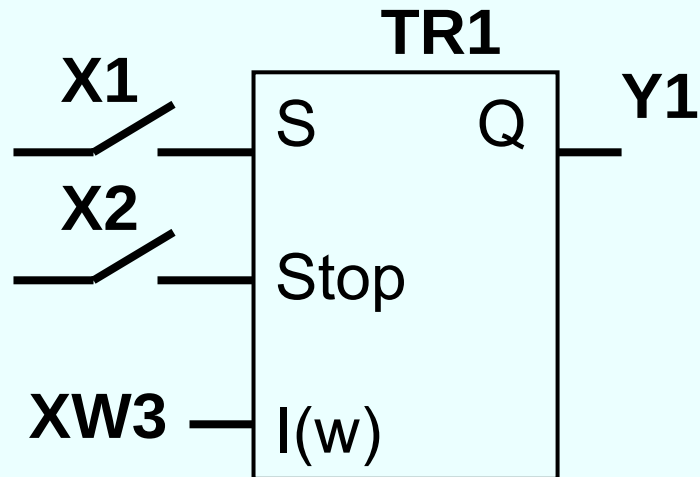
[]Stop: **X2**

[w]I: **KW 100**

[]Q: **Y1**



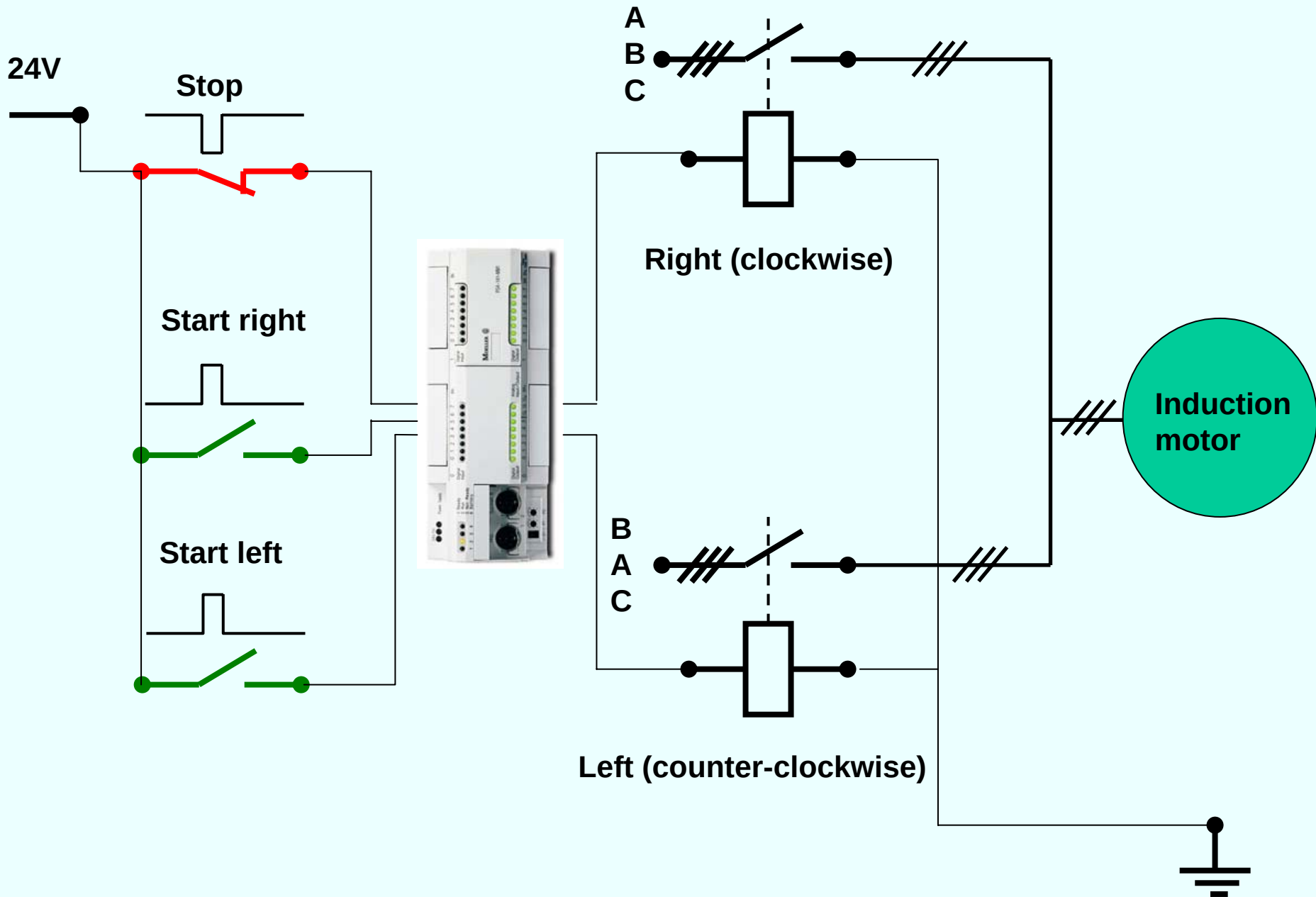
A set of such blocks...



...that can be freely *programmed*
to give *logic*
needed to *control* a given process...

...makes a PLC
(=Programmed Logic Cotroller,
Sterownik Swobodnie Programowalny).





Exercise 2

You've got:

2 relays with 3-f main contacts and WITHOUT auxiliary contacts

2 push button switches NO (green ones)

1 push button switch NC (red one)

1 induction motor

Your goal is to operate drive in two directions safely (without a risk of short circuiting the mains). Write an appropriate code for a PLC from previous slide.

Let the competition continue!